

# Vikas Kumar, Ph.D.

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## Bio

I completed my Ph.D. from the Department of Mathematics, Indian Institute of Technology Roorkee, under the supervision of Prof. Aditi Gangopadhyay. During my Ph.D., I worked jointly at the Quantum-Safe Designs and Analysis Lab (QSDAL), led by Prof. Sugata Gangopadhyay, in the Department of Computer Science and Engineering, IIT Roorkee. Currently, I am working as a postdoctoral researcher in the Department of Computer Science at the University of Calgary, under the supervision of Prof. Reihaneh Safavi-Naini.

My research focuses on the design and cryptanalysis of lattice-based post-quantum cryptosystems, with a particular emphasis on NTRU-like constructions. I am also interested in other cryptographic primitives, such as digital signatures and their applications.

## Positions

|                                   |                                                                                                                                                             |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11 January 2026 – Present         | 📌 Postdoctoral Associate at the Department of Computer Science, University of Calgary, Canada.                                                              |
| 15 August 2025 – 30 November 2025 | 📌 Postdoctoral researcher at Information Security and Machine Learning Lab, under the supervision of Prof. Seong Oun Hwang, Gachon University, South Korea. |
| 22 May 2025 – 28 June 2025        | 📌 Project associate at QSDAL, Department of Computer Science and Engineering, IIT Roorkee, India.                                                           |

## Education

|                 |                                                                                                                                                                   |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2020 – May 2025 | 📌 <b>Ph.D.</b> , Indian Institute of Technology Roorkee<br>Thesis title: <i>Design and Cryptanalysis of Noncommutative NTRU-like Post-Quantum Cryptosystems</i> . |
| 2018 – 2020     | 📌 <b>M.Sc. Mathematics</b> with GPA 9.07, Indian Institute of Technology Bombay                                                                                   |
| 2015 – 2018     | 📌 <b>B.Sc.(Honors) Mathematics</b> with GPA 9.203, Shaheed Bhagat Singh College (University of Delhi)                                                             |
| 2015            | 📌 <b>Intermediate</b> with 94.2%, Sacred Heart Convent School Chandausi                                                                                           |
| 2013            | 📌 <b>High School</b> with 92.5%, Sacred Heart Convent School Chandausi                                                                                            |

## Publications

### Journal Articles

- 1 V. Kumar, A. Raya, S. Gangopadhyay, and A. K. Gangopadhyay, "Cryptanalysis of Group Ring NTRU: The Case of the Dihedral Group," *Security and Privacy*, vol. 8, pp. 1–15, 2025. 🔗 URL: <https://doi.org/10.1002/spy2.70020>.
- 2 A. Raya, V. Kumar, S. Gangopadhyay, and A. Kar Gangopadhyay, "Efficient key encapsulation mechanisms from noncommutative NTRU," *Computer Networks*, vol. 272, p. 111704, 2025. 🔗 DOI: <https://doi.org/10.1016/j.comnet.2025.111704>.
- 3 V. Kumar, R. Das, and A. K. Gangopadhyay, "GR-NTRU: Dihedral group over ring of Eisenstein integers," *Journal of Information Security and Applications*, vol. 83, p. 103795, 2024. 🔗 URL: <https://doi.org/10.1016/j.jisa.2024.103795>.

- 4 A. K. Gangopadhyay, V. Kumar, P. Stănică, and S. Gangopadhyay, "Stability of the Walsh–Hadamard spectrum of cryptographic Boolean functions with biased inputs," *Journal of Applied Mathematics and Computing*, vol. 69, pp. 3337–3357, 2023.  URL: <https://doi.org/10.1007/s12190-023-01887-3>.
- 5 V. Kumar, B. Mandal, and A. K. Gangopadhyay, "On the Gowers  $U_2$  and  $U_3$  norms of Boolean functions and their restriction to hyperplanes," *Discrete Applied Mathematics*, vol. 341, pp. 4–8, 2023.  URL: <https://doi.org/10.1016/j.dam.2023.07.024>.

## Conference Proceedings

- 1 A. Raya, V. Kumar, A. K. Gangopadhyay, and S. Gangopadhyay, "Giant Does NOT Mean Strong: Cryptanalysis of BQTRU," in *Post-Quantum Cryptography - PQCrypto 2025*, R. Niederhagen and M.-J. O. Saarinen, Eds., Cham: Springer Nature Switzerland, 2025, pp. 312–348.  DOI: [10.1007/978-3-031-86599-2\\_11](https://doi.org/10.1007/978-3-031-86599-2_11).
- 2 V. Kumar, A. Raya, A. K. Gangopadhyay, and S. Gangopadhyay, "Dimension reduction attack on noncommutative group ring NTRU over the dihedral group," in *2024 1st International Conference On Cryptography And Information Security (VCRIS)*, vol. 1, 2024, pp. 1–6.  DOI: [10.1109/VCRIS63677.2024.10813443](https://doi.org/10.1109/VCRIS63677.2024.10813443).
- 3 V. Kumar, A. Raya, A. K. Gangopadhyay, S. Gangopadhyay, and M. T. Hussain, "An Efficient Noncommutative NTRU from Semidirect Product," in *Progress in Cryptology – INDOCRYPT 2024*, Springer Nature Switzerland, 2024, pp. 3–27.  URL: [https://doi.org/10.1007/978-3-031-80308-6\\_1](https://doi.org/10.1007/978-3-031-80308-6_1).
- 4 A. Raya, V. Kumar, and S. Gangopadhyay, "DiTRU: A Resurrection of NTRU over Dihedral Group," in *Progress in Cryptology - AFRICACRYPT 2024*, Springer Nature Switzerland, 2024, pp. 349–375.  URL: [https://doi.org/10.1007/978-3-031-64381-1\\_16](https://doi.org/10.1007/978-3-031-64381-1_16).
- 5 A. Raya, V. Kumar, S. Gangopadhyay, and A. K. Gangopadhyay, "Results on the Key Space of Group-Ring NTRU: The Case of the Dihedral Group," in *Security, Privacy, and Applied Cryptography Engineering*, Springer Nature Switzerland, 2024, pp. 1–19.  URL: [https://doi.org/10.1007/978-3-031-51583-5\\_1](https://doi.org/10.1007/978-3-031-51583-5_1).
- 6 V. Kumar, B. Mandal, A. K. Gangopadhyay, and S. Gangopadhyay, "Computational Results on Gowers  $U_2$  and  $U_3$  Norms of Known S-Boxes," in *Codes, Cryptology and Information Security*, Springer Nature Switzerland, 2023, pp. 150–157.  URL: [https://doi.org/10.1007/978-3-031-33017-9\\_10](https://doi.org/10.1007/978-3-031-33017-9_10).

## Preprints

- 1 A. Raya, V. Kumar, K. Dey, and S. Gangopadhyay, *CoNAN: A structure-aware framework for lattice cryptanalysis*, Cryptology ePrint Archive, Paper 2026/1041, 2026.  URL: <https://eprint.iacr.org/2026/1041>.
- 2 A. Raya, V. Kumar, S. O. Hwang, and S. Gangopadhyay, *Almost NTRU: Revisiting Noncommutativity Against Lattice Attacks*, Cryptology ePrint Archive, Paper 2025/1988, 2025.  URL: <https://eprint.iacr.org/2025/1988>.

## Skills

Languages  English, Hindi.  
 Coding  Python, Sagemath (basics)

## Teaching Assistantship

 MAN-001 Linear Algebra and Calculus at IIT Roorkee.

## Teaching Assistantship (continued)

- MAN-006 Probability and Statistics at IIT Roorkee.
- Linear Algebra and Statistics in Course Era Data Science and Machine Learning Course in collaboration with IIT Roorkee.
- Mentored postgraduate students on cryptography projects under my Ph.D. supervisor.
- Assisted Ph.D. students with their pre-Ph.D. coursework in cryptography.

## Miscellaneous

### Awards

- |           |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2020      | <ul style="list-style-type: none"><li>Institute Silver Medal for securing the highest marks in M.Sc. Mathematics, IIT Bombay.</li><li>Prof. P.V. Sukhatme Memorial Prize for being in the top two M.Sc. students (in terms of GPA) in the graduating batch of the Mathematics programme, IIT Bombay.</li><li>Mrs. Rama Mathur Memorial Prize for securing the highest CPI in M.Sc. Mathematics, IIT Bombay.</li></ul> |
| 2018      | <ul style="list-style-type: none"><li>Gold medal for securing first rank in B.Sc. (Hons.) Mathematics in Shaheed Bhagat Singh College, University of Delhi.</li></ul>                                                                                                                                                                                                                                                 |
| 2015–2018 | <ul style="list-style-type: none"><li>Awarded Kamakshi Trehan Memorial Merit Scholarship for securing first rank in I, II, III and VI semesters in B.Sc. (Hons.) Mathematics in Shaheed Bhagat Singh College, University of Delhi.</li></ul>                                                                                                                                                                          |

### Achievements

- |      |                                                                                                                                                       |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2020 | <ul style="list-style-type: none"><li>All India Rank 125 in GATE Mathematics exam.</li></ul>                                                          |
| 2019 | <ul style="list-style-type: none"><li>All India Rank 55 in CSIR NET Mathematics exam</li></ul>                                                        |
| 2018 | <ul style="list-style-type: none"><li>All India Rank 42 in CSIR NET Mathematics exam</li><li>All India Rank 20 in IIT JAM Mathematics exam.</li></ul> |
| 2015 | <ul style="list-style-type: none"><li>Overall topper in school in Intermediate.</li></ul>                                                             |
| 2013 | <ul style="list-style-type: none"><li>Secured the second-highest marks in class in High School.</li></ul>                                             |

### Conferences/Workshops attended

- |               |                                                                                                                                                                                                                                                                                                                                                                        |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| December 2025 | <ul style="list-style-type: none"><li>26th International Conference on Cryptology in India (INDOCRYPT 2025), Bhubaneswar, India.</li></ul>                                                                                                                                                                                                                             |
| April 2025    | <ul style="list-style-type: none"><li>16th International Conference on Post-Quantum Cryptography. (PQCRYPTO 2025), Academia Sinica, Taipei, Taiwan.</li></ul>                                                                                                                                                                                                          |
| February 2025 | <ul style="list-style-type: none"><li>Workshop on Advanced Topics in Trusted Information Computing (ATTIC), IITKGP, Kharagpur, India.</li></ul>                                                                                                                                                                                                                        |
| December 2024 | <ul style="list-style-type: none"><li>25th International Conference on Cryptology in India (INDOCRYPT 2024), Chennai, India.</li><li>Short-term course on current topics in Cyber Security, IIT Roorkee, India.</li><li>30th International Conference on the Theory and Application of Cryptology and Information Security (ASIACRYPT 2024), Kolkata, India.</li></ul> |
| April 2024    | <ul style="list-style-type: none"><li>Workshop on Lattice-based Post-quantum Cryptography 2024, Department of Computer Science, Ashoka University, India.</li></ul>                                                                                                                                                                                                    |

## Miscellaneous (continued)

- March 2024  22nd International Conference on Applied Cryptography and Network Security (ACNS 2024), NYU Abu Dhabi, UAE.
- December 2023  13th International Conference on Security, Privacy and Applied Cryptographic Engineering 2023 (SPACE 2023), IIT Roorkee, India.
- December 2022  12th International Conference on Security, Privacy and Applied Cryptographic Engineering 2022 (SPACE 2022), LNMIIT Jaipur, India.

## Talks

- June 2025  Invited as a guest lecturer on Post-Quantum and Lattice-Based Cryptography for the short-term course “Computer and Network Security in the Post-Quantum Era” at Sharda University, Greater Noida, India.
- March 2025  Mentored and trained participants in the faculty development program on “Cyber security: present and future” organized by the Department of Computer Science and Engineering, IIT Roorkee.
- December 2024  Lectured on Lattices and their relation with NTRU at a boot camp on "Future Security Technologies and Hardware Design" organized by the Department of Computer Science and Engineering, IIT Roorkee.

## Professional Services

-  Subreviewer for the conferences INDOCRYPT 2024.
-  Subreviewer for the conferences SPACE 2023.
-  Reviewer for the journal Discrete Applied Mathematics.

## References

### Prof. Aditi Kar Gangopadhyay

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